



上海科技大学
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EE115 Analog Circuits 模拟电路

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Outline



- Semiconductors
- SEDTRA/SMITH book page 135-148;

Silicon: Group IV Element

- Why silicon is selected to fabricate IC?

IA	IIA	IIIB	IVB	VB	VIB	VII B	VIII	IB	IIB	IIIA	IVA	VA	VIA	VIIA	GASES		
1 H 1.00797															2 He 4.0026		
3 Li 6.939	4 Be 9.0122																
11 Na 22.9898	12 Mg 24.312																
19 K 39.102	20 Ca 40.08	21 Sc 44.956	22 Ti 47.90	23 V 50.942	24 Cr 51.996	25 Mn 54.9380	26 Fe 55.847	27 Co 58.9332	28 Ni 58.71	29 Cu 63.54	30 Zn 65.37	31 Ga 69.72	32 Ge 72.59	33 As 74.9216	34 Se 78.96	35 Br 79.909	36 Kr 83.80
37 Rb 85.47	38 Sr 87.62	39 Y 88.905	40 Zr 91.22	41 Nb 92.906	42 Mo 95.94	43 Tc (99)	44 Ru 101.07	45 Rh 102.905	46 Pd 106.4	47 Ag 107.870	48 Cd 112.40	49 In 114.82	50 Sn 118.69	51 Sb 121.75	52 Te 127.60	53 I 126.904	54 Xe 131.30
55 Cs 132.905	56 Ba 137.34	*57 La 138.91	72 Hf 178.49	73 Ta 180.948	74 W 183.85	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.09	79 Au 196.967	80 Hg 200.59	81 Tl 204.37	82 Pb 207.19	83 Bi 208.980	84 Po (210)	85 At (210)	86 Rn (222)
87 Fr (223)	88 Ra (226)	†89 Ac (227)	104 Rf (261)	105 Db (262)	106 Sg (266)	107 Bh (262)	108 Hs (265)	109 Mt (266)	110 ? (271)	111 ? (272)	112 ? (277)						

P-type dopant

N-type dopant

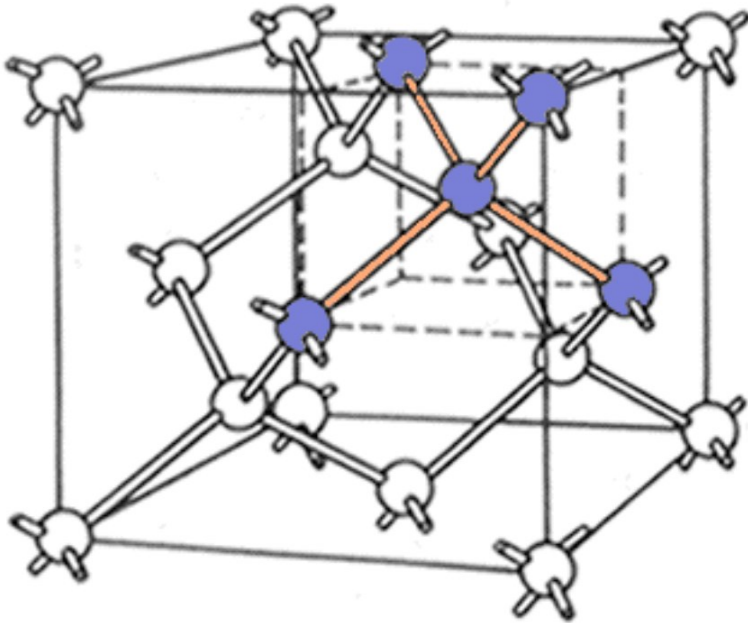
P-type
dopant

N-type
dopant

Silicon



Crystalline Structure (Diamond Cubic)



2-D Representation

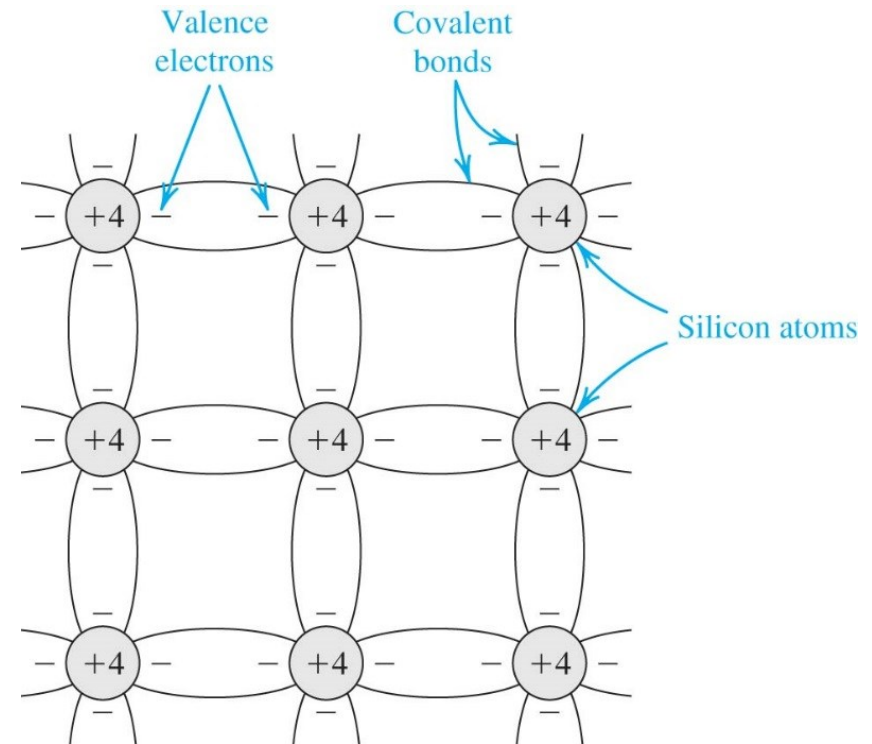


Figure 3.1.

At 0 Kelvin, all electrons are “locked” in
covalent bonds (共价键)

-> Behave like insulator (绝缘体)

Electrons and Holes

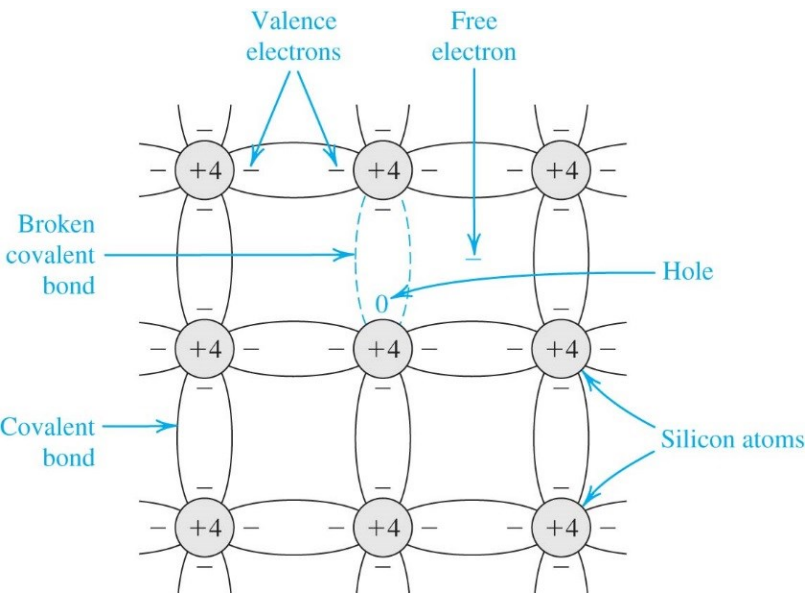


Figure 3.2

- **Thermal Generation:**
 - At room temperature, thermal energy breaks some covalent bonds, creating free electrons and “holes”
- **Hole (空穴):** empty space left by electron
 - Hole “**moves**” as adjacent electron move into its space
 - Treat hole like a positively charged particle
- **Recombination (复合)**
 - Free electrons and holes move randomly through the silicon crystal structure
 - Some electrons may fill some of the holes

Intrinsic (本征) Semiconductor

- Intrinsic semiconductor

- $n = p = n_i$
- n : electron concentration [cm^{-3}]
- p : hole concentration [cm^{-3}]
- $n_i = BT^{3/2}e^{-E_g/(2kT)}$: intrinsic carrier concentration
 - B : material dependent constant
 - T : temperature in Kelvin
 - E_g : **bandgap(帶隙)** energy (=1.12 eV for Si)
 - k : Boltzmann's constant = 8.62×10^{-5} eV/K
- At room temperature ($T = 300 \text{ K}$)

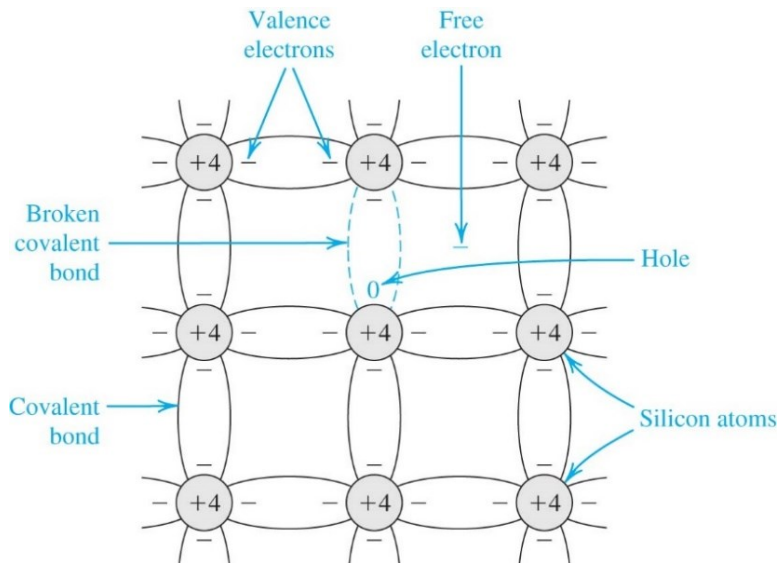


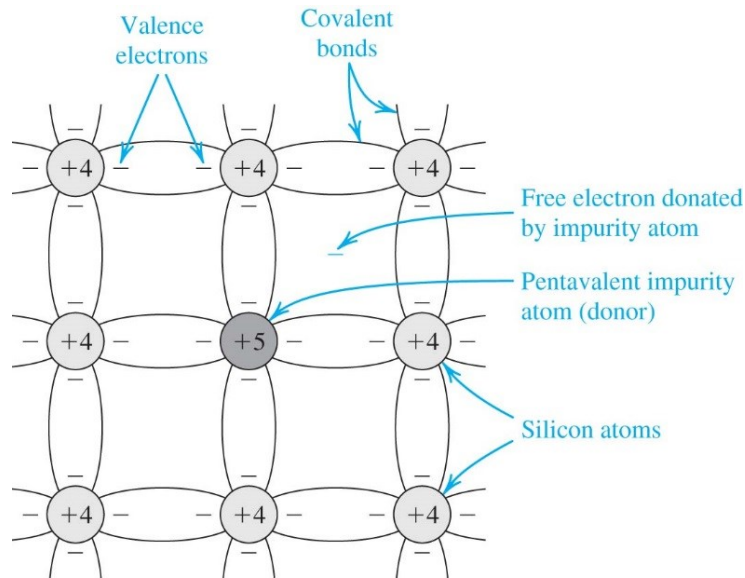
Figure 3.2

$$n_i = 1.5 \times 10^{10} [\text{cm}^{-3}]$$

Note: There are 5×10^{22} atoms/ cm^{-3} , so the number of free electrons and holes are very small

In general, $np = n_i^2$

N-Type Semiconductor



- Electron concentration can be greatly increased by replacing some Si atoms with P (phosphorus) or As (Arsenic), which have 5 shell electrons (one more than Si).
- P or As are called “**donors (施主)**”
 $n_n = N_D$ (donor impurity concentration)
 $p_n = n_i^2 / N_D$ where $n_i = 1.5 \times 10^{10} [\text{cm}^{-3}]$
- Subscript n refers to n-type semiconductor (n stands for "negative", referring to the charge carried by electrons)
- In n-type semiconductor, $n_n \gg n_i \gg p_n$
- e.g., $N_D = 10^{17} \text{ cm}^{-3}$, $n_n = 10^{17}$, $p_n = 2.2 \times 10^3$
- Electrons are “**majority carriers (多子)**”
- Holes are “**minority carriers (少子)**”

Figure 3.3 A silicon crystal doped by a pentavalent element. Each dopant atom donates a free electron and is thus called a donor. The doped semiconductor becomes n type.

P-Type Semiconductor

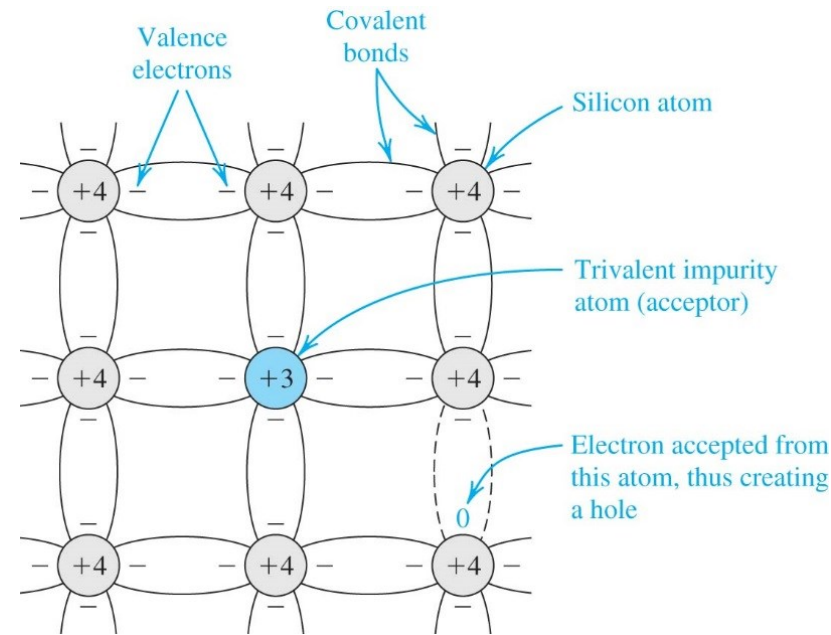


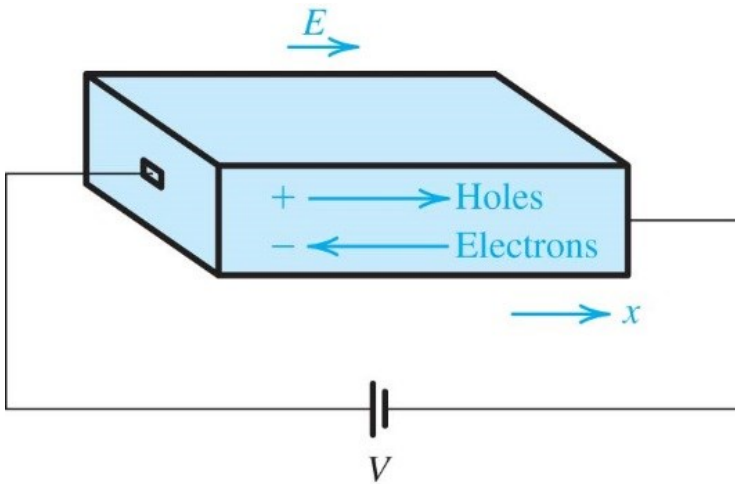
Figure 3.4 A silicon crystal doped with boron, a trivalent impurity. Each dopant atom gives rise to a hole, and the semiconductor becomes *p* type.

- Hole concentration can be greatly increased by replacing some Si atoms with B (boron), which has 3 shell electrons (one less than Si).
- B is called “**acceptors**” (受主)
 $p_p = N_A$ (acceptor impurity concentration)
 $n_p = n_i^2 / N_A$ where $n_i = 1.5 \times 10^{10} [\text{cm}^{-3}]$
- Subscript *p* refers to p-type semiconductor (*p* stands for “positive”, referring to the charge carried by holes)
- In p-type semiconductor, $p_p \gg n_i \gg n_p$
- e.g., $N_A = 10^{17} \text{ cm}^{-3}$, $p_p = 10^{17}$, $n_p = 2.2 \times 10^3$
- Holes are “majority” carriers
- Electrons are “minority” carriers

Current in Semiconductor (1): Drift Current (漂移电流)



When an **electrical field, E** , is applied, holes moves in the direction of E , while electrons move opposite to E :



$$\begin{cases} v_{p-drift} = \mu_p E \\ v_{n-drift} = -\mu_n E \end{cases}$$

Where, μ_p is the **hole mobility (空穴迁移率)**; μ_n is the **electron mobility (电子迁移率)**;

In intrinsic Si, $\mu_n = 1350 \text{ cm}^2/\text{V}\cdot\text{s}$

$$\mu_p = 480 \text{ cm}^2/\text{V}\cdot\text{s} \quad (\text{Note: } \mu_n \approx 2.5\mu_p)$$

Current density, J [A/cm²]

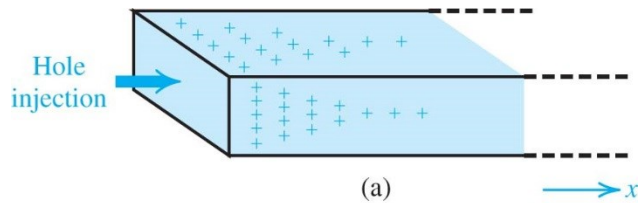
$$J = qp v_{p-drift} + qn v_{n-drift} = q(p\mu_p + n\mu_n)E = \sigma E$$

where $\sigma = q(p\mu_p + n\mu_n)$ is **conductivity (电导率)** [S/cm]

Resistivity (电阻率) $\rho = 1/\sigma$ [Ω-cm]

Figure 3.5 An electric field E established in a bar of silicon causes the holes to drift in the direction of E and the free electrons to drift in the opposite direction. Both the hole and electron drift currents are in the direction of E .

Current in Semiconductor (2) Diffusion Current (扩散电流)

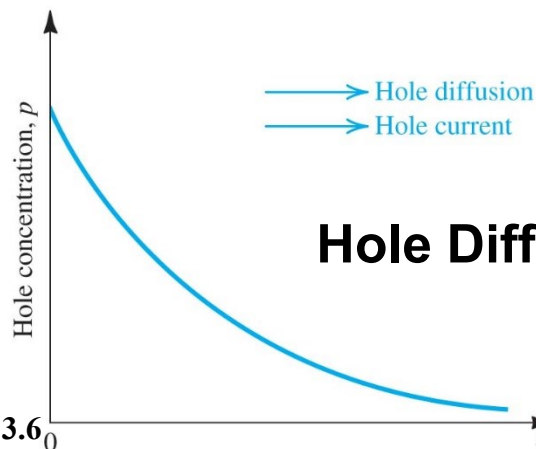


Diffusion: particles move from high to low concentrations. As electrons and holes diffuse, currents flow because they carry charges.

$$J_{p-diff} = -qD_p \frac{dp(x)}{dx}$$

$$J_{n-diff} = -(-q)D_n \frac{dn(x)}{dx} = qD_n \frac{dn(x)}{dx}$$

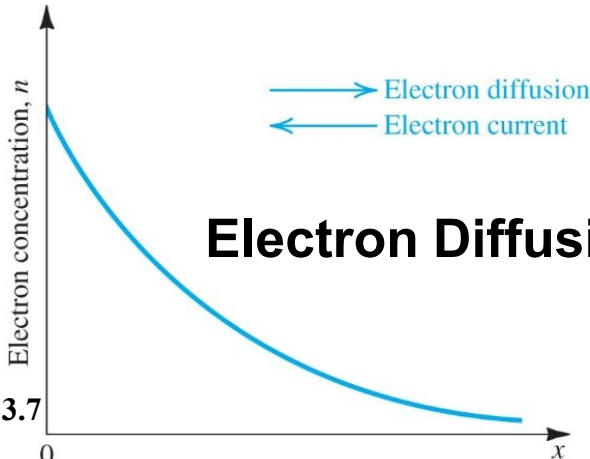
Hole Diffusion



where D_p and D_n are hole and electron **diffusion coefficients/diffusivity (扩散率)** [cm^2/s]

In intrinsic Si, $D_p = 12 \text{ cm}^2/\text{s}$, $D_n = 35 \text{ cm}^2/\text{s}$
Total diffusion current density, J_{diff} [A/cm^2]

Electron Diffusion



$$J_{diff} = -qD_p \frac{dp(x)}{dx} + qD_n \frac{dn(x)}{dx}$$



Example

- Given a silicon bar with hole concentration profile

$$p(x) = p_0 e^{-x/L_p}$$

Find the hole-current density at $x = 0$. Let $p_0 = 10^{16}/\text{cm}^3$, $L_p = 1 \text{ } \mu\text{m}$, and $D_p = 12 \text{ cm}^2/\text{s}$.

If the cross-sectional area of the bar is $100 \text{ } \mu\text{m}^2$, find the current I_p .

Solution

$$\begin{aligned} J_p &= -qD_p \frac{dp(x)}{dx} \\ &= -qD_p \frac{d}{dx} [p_0 e^{-x/L_p}] \\ &= q \frac{D_p}{L_p} p_0 e^{-x/L_p} \end{aligned}$$

Thus,

$$\begin{aligned} J_p(0) &= q \frac{D_p}{L_p} p_0 \\ &= 1.6 \times 10^{-19} \times \frac{12}{1 \times 10^{-4}} \times 10^{16} \\ &= 192 \text{ A/cm}^2 \end{aligned}$$

The current I_p can be found from

$$\begin{aligned} I_p &= J_p \times A \\ &= 192 \times 100 \times 10^{-8} \\ &= 192 \text{ } \mu\text{A} \end{aligned}$$

Einstein Relationship



$$\frac{D_n}{\mu_n} = \frac{D_p}{\mu_p} = V_T = \frac{kT}{q}$$

- V_T : Thermal voltage. At room temperature, $V_T = 26$ mV

Mass transfer by diffusion: where c is concentration

$$J_D = -D \frac{\partial c}{\partial x}$$

- Drift mass transfer

$$J_F = \mu c F \quad F = -\partial V / \partial x$$

- In equilibrium, total current is zero $\mu c F - D \frac{\partial c}{\partial x} = 0 \quad c \sim \exp[-V(x)/k_B T]$

$$D = k_B T \mu$$

Homework2-due March 31st 11p.m.



SEDRA/SMITH Book chapter 2 problem 2.95,
2.97, 2.109, 2.112; chapter 3 problem 3.2, 3.3,
3.8